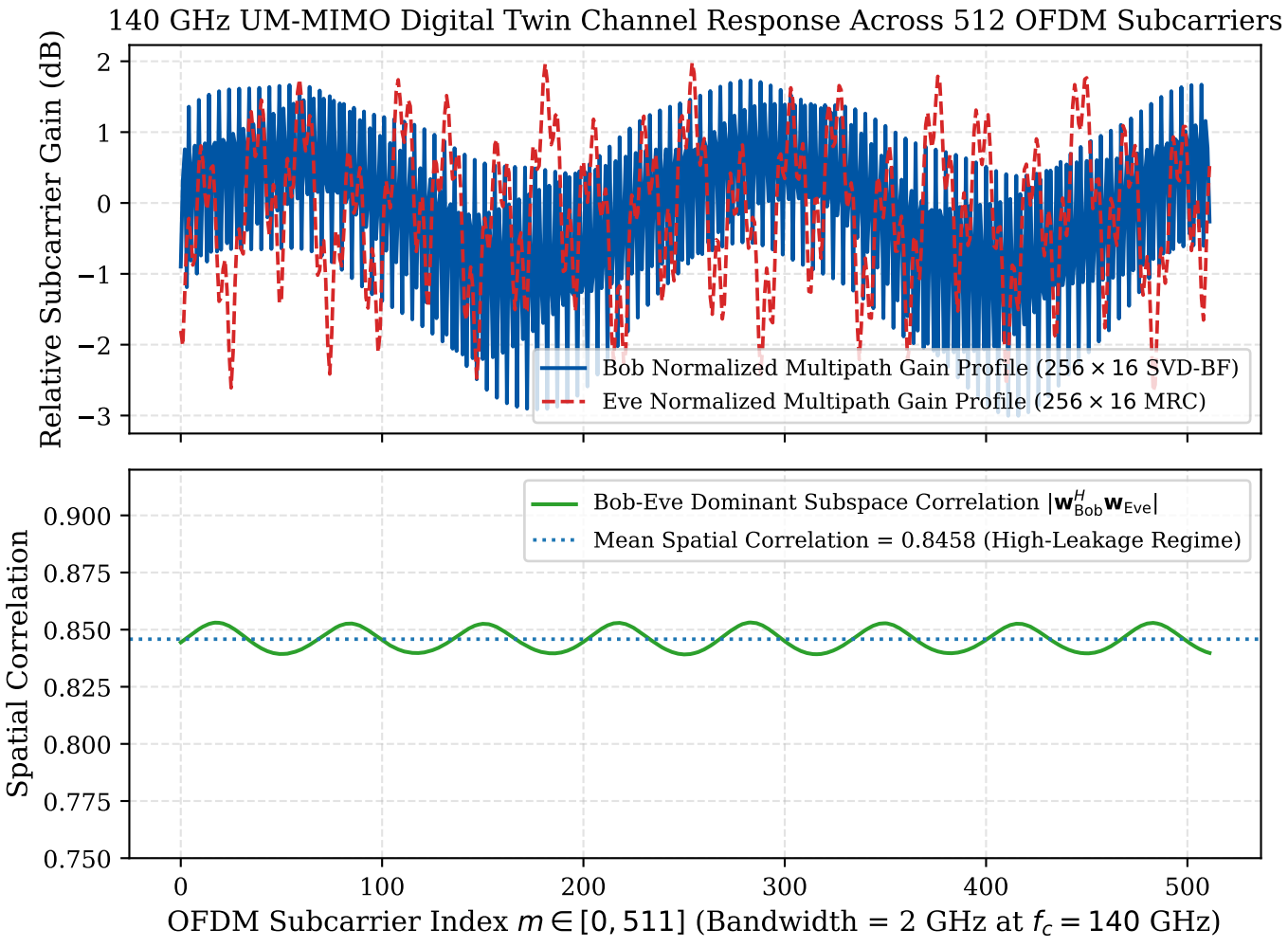
140 GHz UM-MIMO Digital Twin Channel Response Across 512 OFDM Subcarriers
Relative Subcarrier Gain (dB)
2
1
0
−1
−2
−3
Bob Normalized Multipath Gain Profile (256 × 16 SVD-BF)
Eve Normalized Multipath Gain Profile (256 × 16 MRC)
Spatial Correlation
0.900
0.875
0.850
0.825
0.800
0.775
0.750
Bob-Eve Dominant Subspace Correlation $|\mathbf{w}_{\text{Bob}}^H \mathbf{w}_{\text{Eve}}|$
Mean Spatial Correlation = 0.8458 (High-Leakage Regime)
0
100
200
300
400
500
OFDM Subcarrier Index $m \in [0, 511]$ (Bandwidth = 2 GHz at $f_c = 140$ GHz)